PHANEENDRA B

Staff Backend Engineer | Technical Architect | 14+ Years Designing Production-Grade Distributed Systems & AI Orchestration Engines

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PROFESSIONAL SUMMARY

Strategic, hands-on **Staff Backend Engineer & Technical Architect** with **14+ years** of professional experience designing production-grade, high-throughput **Microservices Architectures** and highly **resilient distributed systems** across enterprise product, banking, and AI orchestration platforms. Proven mastery engineering **stateful task execution layers, asynchronous job queues, and fault-tolerant multi-agent topologies** using **Java, Python, FastAPI, Spring Boot, Kafka, Temporal, Qdrant, MongoDB and PostgreSQL**. Expert in balancing core system design constraints—such as high availability, strict data consistency, and low-latency **p99 performance profiles**—with cutting-edge **LLMOps and FinOps infrastructure**, managing the complete technical lifecycle from initial system architecture to global production scale.

ARCHITECTURE HIGHLIGHTS

- **Award-Winning Distributed Infrastructure:** Designed the high-throughput, event-driven distributed infrastructure powering Floik's core engine, achieving **#1 Product of the Day**, **#1 Product of the Week**, and winning Product Hunt's **2023 Golden Kitty Award**.
- **High-Throughput Concurrency Scale:** Scaled media processing and ingestion pipelines to handle **10K+ concurrent records/day**, lowering **p99 processing latency by 30%** through precise chunk-based processing configurations, database routing, and bounded async concurrency loops.
- **Deterministic Distributed State Machines:** Deployed highly **resilient, fault-tolerant execution pipelines** utilizing **LangGraph and Temporal**, establishing strict data idempotency patterns and stateful orchestration layers that guarantee **100% state recovery** under extreme traffic surges.
- **Secure Enterprise API Gateways:** Architected production-grade **Model Context Protocol (MCP)** servers alongside custom middleware interceptors to expose monolithic core APIs as strongly-typed, secure tools, enforcing runtime compliance via **PII Detection, Prompt Injection Defence, and Human-In-The-Loop (HITL) Verification gates**.
- **High-Scalability Vector Indexing:** Optimized distributed vector storage and advanced **RAG search frameworks**, increasing autonomous task accuracy by **25%** using high-scalability **Qdrant vector partitioning, metadata filtering**, and **PostgreSQL analytics** lookup loops.

TECHNICAL SKILLS

Architecture & Core Design: System Design, Microservices Architecture, Event-Driven Architecture, High-Availability Distributed Systems, Fault-Tolerant Topology, Async Job Queues, Task Planners, DAG Engines, CAP Theorem Zoning
Enterprise AI & Orchestration: Agentic Systems, Multi-Agent Topologies, State Management, Memory Engineering (Sliding Window, Episodic Memory), Model Context Protocol (MCP), LangGraph, LlamaIndex, LangChain
LLMOps & FinOps Infrastructure: Model Routers, Multi-Provider Fallback Proxies, Semantic Caching (RedisVL), Context Window Optimization, Multi-Tenant Token Budgeting, Rate-Limiting Cost Allocation, RAG Evaluations (Ragas, Context Recall, Faithfulness)
Languages & Frameworks: Java (8/17/21), **Python**, Spring Boot, **FastAPI**, React (TypeScript), Core Frameworks & SDKs
Data, Messaging & Vector: Apache Kafka, ZMQ, Redis, Qdrant (Vector DB Architecture), PostgreSQL (Relational Analytics), MySQL, MongoDB, Elasticsearch, Change Data Capture (CDC Pipeline Orchestration)
Cloud, DevOps & Infrastructure: AWS (EKS, Lambda, S3, MSK, CloudFront, EventBridge, KMS, MediaConvert), GCP, Docker, Kubernetes, CI/CD Automation (Jenkins, GitHub Actions), SonarQube, JVM Concurrency Profiling

EXPERIENCE

Kovai Systems India Private Limited, Bangalore — Technical Lead Architect / SDE-3 (Architect Track)

Nov 2024 – Present

Kovai Systems acquired Floik, the product built at Ping Technologies — same product and team, under a new parent company.

***Projects:** Floik, Document360 Integration*

- Led an agile team of **5 engineers** to architect a secure, distributed **Multi-Agent Topology**; built highly **resilient, horizontally scalable microservices** using **Java, Python, FasiAPI, Spring Boot, Kafka, and Temporal** (Infrastructure Orchestrator) to handle stateful task planning, **Human-In-The-Loop (HITL) Verification gates**, and deterministic recovery of production workflows.

- Engineered advanced enterprise **RAG & Ingestion Strategies** featuring **Recursive Character Chunking** and syntax-aware code parsing; implemented a high-performance **Retrieval & Reranking Architecture** combining **Dense Vector Search (Qdrant)** and **Sparse Keyword Search (BM25)** aligned via **Reciprocal Rank Fusion (RRF)**, validating pipeline performance using the **Ragas framework** (**Context Recall & Faithfulness metrics**).
- Architected a real-time **AI Data Pipeline Orchestration** layer utilizing **Change Data Capture (CDC)** to auto-embed and sync high-volume operational databases into partitioned **Qdrant** collections with strict data isolation, structural metadata filtering, and optimized replication factors.
- Maintained **p99 processing latencies under strict baselines** by managing media processing pipelines processing **10K+ concurrent video tasks daily**; leveraged fine-tuned database routing and bounded parallel execution loops to handle sudden infrastructure load shifts.
- Acted as cross-functional anchor, unblocking frontend architectural milestones during resource constraints by temporarily operating as a full-stack **React** engineer to deploy, test, and ship crucial application views.

Ping Technologies, Bangalore — Technical Lead Architect / SDE-3 (Architect Track)

Jan 2022 – Nov 2024

Projects: *Floik (storytelling platform generating visually rich interactive product demos, videos, and step-by-step guides)*

- Contributed core platform engineering and high-availability architecture for Floik's high-visibility launch, maintaining **fault-tolerant system health** to capture the **2023 Golden Kitty Award** and **#1 Product of the Day/Week** validation.
- Architected an intelligent **LLMOps Model Router & Fallback proxy** using python across multi-provider cloud layers (OpenAI/Anthropic/Azure); integrated a high-speed **Semantic Caching Engine (RedisVL)** to catch redundant prompts, achieving **instantaneous failover boundaries** and slashing API token bills.
- Engineered secure, high-throughput RESTful gateways embedded with custom **FinOps Cost Allocation tables** , multi-tenant token tracking, **Context Window Optimization** (dynamic history trimming), and token-bucket **Rate Limiting** to seamlessly govern shared enterprise API keys.
- Guaranteed a **99.99% service availability baseline** by designing resilient event-driven microservices anchored on **CAP theorem principles, active failure-isolation zones, and strict data idempotency patterns**.
- Executed comprehensive **Python** application profiling and runtime optimization using **cProfile** and **Py-Spy**; eliminated CPU and memory bottlenecks by decomposing routines into isolated worker pools via the multiprocessing API, shifting I/O tasks to non-blocking asyncio loops utilizing uvloop, and fine-tuning reference collection loops (gc thresholds) to stabilize **p99 metrics under heavy processing loads**.
- Designed highly resilient, event-driven pipelines on AWS using **Lambda**, **Step Functions**, **EventBridge**, and **Kafka**, embedding dead letter queues, automated retry behavior, and distributed tracing from initial draft to production rollout.
- Established comprehensive testing infrastructure (JUnit, Mockito, and automated contract testing blocks) paired with automated DevOps execution (**Kubernetes, AWS EKS, Jenkins, GitHub Actions, and SonarQube quality gates**) to protect core builds against regression errors.

Razorthink (Technologies & Software), Bangalore — Software Developer -> SDE-3 / Senior Backend Developer

Jun 2012 – Jan 2022

Projects: *Razorthink aiOS, FormAI, Customer Churn, Next Best Product, Demand Forecasting, Adonme, Anounze, Guest Verification Portal, Incident Management System, OurUpgrade, SafelyStay*

- Directed data validation, analysis, and preprocessing architecture for Customer Churn, Demand Forecasting, and Next Best Product initiatives in the banking sector.
- Designed a multi-module, large-scale data processing engine and AI platform supporting batch and real-time processing; constructed automated **DAG infrastructure and data-ingestion task planners** to optimize heavy asynchronous pipelines.
- Defined process improvements that reduced data-preprocessing cycle times.
- Performed **Java performance tuning and JVM profiling**, optimizing **concurrency, multithreading, garbage collection (GC) JVM configurations, Python Asyncio Coordination & Process Pooling** and **Heuristic Memory Auditing** to identify and resolve CPU, memory, thread, and latency bottlenecks and improve system throughput.
- Enforced rigorous code quality via comprehensive JUnit, Mockito, and contract testing layers to maximize regression coverage.

EDUCATION

Bachelor of Technology in **Computer Science and Engineering** (B.Tech CSE)
— Sri Sivani College of Engineering, JNTU-Kakinada, 2007 - 2011